PROJECT ABSTRACT

Modelling the Progression of Breast Cancer

Using Data-Driven & Machine Learning Techniques

Group Member	Student Number
Siyabonga Dunn	223017814
Sfiso Gumede	223063631
Alunge Hlubi	223019985

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Abstract

This project focuses on modelling the progression of breast cancer using data-driven sequence modelling and machine learning techniques. Breast cancer remains one of the leading causes of oncology-related morbidity worldwide, where capturing dynamic progression trajectories and staging shifts is critical for optimizing patient outcomes. The aim of this study is to formally define and evaluate patient clinical histories as sequential trajectories to describe how breast cancer progresses through distinct clinical stages over time. The general approach will involve building a disease progression model that captures sequential transitions between clinical health states (Benign through Stage IV) and predicts progression pathways. To directly align with the core sequence modelling paradigms of this module, the study will implement and compare traditional Markov Chains against Hidden Markov Models (HMMs), the latter being utilized to capture latent, unobserved disease progression states from observable clinical data. Additionally, deep sequence learning architectures, specifically Bidirectional Recurrent Neural Networks (RNNs) like LSTM and GRU with Attention mechanisms, will be explored and benchmarked to evaluate their capacity for capturing long-term dependencies in patient histories compared to traditional state-transition approaches.

The project will utilize a structured longitudinal clinical dataset containing sequential patient records. The data requirements include:

- Patient Sequences: Longitudinal tracking records containing unique patient IDs, chronological year markers, diagnostic labels, and five-class clinical stages.
- Tumor Metrics & References: Thirty numeric cytological features from fine needle aspirate images alongside real-world Wisconsin reference datasets for bias analysis.
- Robustness Testing: Controlled injection of 0% to 20% label noise into test sequences to evaluate model degradation.

Proposed Timeline:

- March 06 – March 20, 2026: Literature review and sequence dataset acquisition
- March 21 – April 10, 2026: Data preprocessing, sequence padding/formatting, and exploratory data analysis
- April 11 – May 02, 2026: Model development (Markov Chains, HMMs, and Attention-driven BiLSTM/GRU baselines)
- May 03 – May 16, 2026: Model evaluation, cross-validation, noise robustness testing, and dataset bias analysis
- May 17 – May 24, 2026: Results analysis, sequence determinism metrics, entropy calculations, and clinical interpretation
- May 25 – June 01, 2026: Final report writing and submission

The expected deliverables include a preprocessed sequential clinical dataset formatted for time-series/state-space analysis; trained and validated Markov Chain, Hidden Markov, LSTM, and GRU sequence models; a comprehensive performance evaluation report detailing state prediction accuracy, log-likelihood, robustness curves, and structural determinism results; and an 8-page final research report detailing the sequence modelling methodology, empirical findings, and clinical recommendations.